

A Formal Comparison of Controllability Definitions for Deterministic and Nondeterministic Automata

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In the field of supervisory control theory the literature often proposes different definitions for the same concept, making it difficult to understand how these definitions are related. This is definitely so for the fundamental notion of controllability of a supervisor w.r.t. a plant. This paper presents an overview of the definitions of controllability found in the literature and studies their relationships in settings of both deterministic and nondeterministic automata. In the general context, where both the supervisor and the plant are allowed to be nondeterministic, the notions of controllability as described by Flordal and Malik, and Σ_u -admissibility by Kushi and Takai are equivalent. These are also the only notions that imply the traditional notion of (language) controllability. From a practical perspective, one is often more interested in controllability of a supervised plant w.r.t. a plant. In this context, in addition to the previous two controllability notions, state controllability by Zhou et al. implies language controllability.

1 Introduction

In the field of supervisory control theory, the basic problem addressed is the synthesis of a supervisor (or supervisory controller) from a model of an uncontrolled system, often referred to as a *plant*, and a specification of the desired behavior of the supervised plant, that is, the plant in closed loop with the supervisor. An alternative formulation is often used where, instead of a specification of desired behavior, a set of requirements is provided that the supervised plant should satisfy. In addition to the above-mentioned satisfaction of a specification or requirements, the synthesis procedure results in a supervisor that is maximally permissive, nonblocking and controllable [19].

A fundamental notion in this problem statement is that of controllability. In supervisory control theory, the behavior of a plant is composed of controllable and uncontrollable events, coarsely representing actuation and sensing in the plant. Intuitively, controllability captures the assumption that a supervisor can only interfere with the controllable events of the plant. Controllability is also of fundamental interest when the supervisor is not obtained by synthesis but by means of another process, e.g., when it is manually crafted.

In the literature, different definitions for controllability have been proposed. It is sometimes hard to understand how these different notions relate and in which context they can be applied. To shed light on this, this paper recalls the definitions of controllability found in the literature and studies their relationships.

Originally, controllability was defined as a relation between a specification language and a plant language [8, 18]. To avoid ambiguity, this notion is called language controllability in the present paper. The purpose is to guarantee that the plant language can be limited (by applying a supervisor) to achieve exactly the specification language by disabling only controllable events in the plant. In other works in

the literature, controllability is also defined directly between a supervisor and a plant. In this case, the purpose is to establish that the supervisor does not disable uncontrollable events in the plant. Finally, definitions of controllability that relate a supervised system and a plant are sometimes found. With language controllability in mind and assuming that the supervisor and the plant have the same set of events, controllability of a supervisor w.r.t. a plant is equivalent to controllability of the supervised plant w.r.t. that plant (see Lemma 1 in Section 4.1).

In the present paper, we prefer to represent both the supervisor and the plant using a (finite) automaton, and we prefer to formulate controllability as a property of a supervisor w.r.t. a plant.

Contribution(s) The contributions of this paper can be summarized as follows:

- We present an overview of the definitions of controllability from the literature and identify the contexts in which they are defined in terms of properties and assumptions on the plant and the supervisor (in Section 4).
- We identify the relations between controllability notions in different contexts of deterministic and nondeterministic plant and supervisor automata, and reflect on the definitions (in Section 5). We determine that only two other definitions imply language controllability in all identified contexts.
- Taking into account the point of view that the controllability of a supervised plant w.r.t. a plant is the more relevant notion, we establish that three notions from the literature are adequate under this interpretation (in Section 6).

2 Related Work

In this paper, we discuss the fundamental notion of controllability in detail. In Table 1 we give an overview of the different controllability notions that we analyze in this paper. We forego a detailed discussion of these notions of controllability here, as they will be studied in detail in Section 4.

Table 1: Overview of controllability notions. Columns S and P indicate whether supervisor and plant are deterministic (D) or nondeterministic (N) automata. Column $\subseteq$ indicates what assumption the corresponding definition makes on the inclusion of languages ($L(S) \subseteq L(P)$, see Definition 2) or automata ($S \subseteq P$, see Definition 4).

Notion	Abbreviation	Definition	S	P	$\subseteq$
Language controllability [8, 18]	LC	5	N	N	
Automata controllability	AC	6	N	N	
Control relation [22]	CR	7	D	D	
State controllability [24]	SC	8	D	N	
Partial bisimulation [22]	PB	9	D	D	
Nondeterministic partial bisimulation [3, 17]	PBⁿ	10	N	N	
Controllability for nondeterministic automata [10, 11]	FM	11	N	N	
Σ_u -admissibility [16]	KT	12	N	N	
Nondeterministic state controllability [26]	SCⁿ	13	N	N	$L(S) \subseteq L(P)$
Controllability for nondeterministic subautomata [10, 11]	FM$\subseteq$	14	N	N	$S \subseteq P$

For a context with both deterministic plant and supervisor, in [2], where a game-based approach for the synthesis of (supervisory) controllers is discussed, a controller is always required to enable any un-

controllable event. In [4] this approach is extended to nondeterministic plants and a state-dependent notion of (un)controllable events. In both works, controllability is captured as a modal μ -calculus formula for which a satisfying model needs to be found. In [21] a method is presented to express controllability in terms of modal μ -calculus formulas such that controllability can be verified using model checking.

Controllability, as a property of a supervised system in comparison with an unsupervised plant, is also relevant in semantically richer contexts. Typically, definitions in such contexts are generalizations of the definitions used in this paper. In the setting with unobservable events, the definition of language controllability is essentially unchanged; see, e.g., [8]. Controllability in a setting with forcible events is considered in [13, 25]. For timed discrete-event systems, controllability is extended in [7]. Generalizations to time-interval discrete-event systems and timed automata with forcible events are given in [6] and [20], respectively.

In some implementations of synthesis (such as in Supremica [1] and ESCET [12]), specifications of admissible behavior by means of automata are dealt with by plantification [11]. In plantification a requirement automaton is made complete by adding all missing uncontrollable transitions with a fresh state as the target state. Safe nonblocking supervisory control can in this way be reduced to nonblocking supervisory control. The result of plantification is controllable w.r.t. each of the notions of controllability studied in this paper.

3 Languages and Automata

Discrete-event systems are used to model systems. In this setting, we distinguish the *plant*, i.e., the system to be controlled, and the *supervisor*. Plants are traditionally modeled using languages [8, 18]. Let Σ be a set of *events*. Then Σ^* denotes the set of all sequences of events. A language L is a subset of Σ^* . We write ε for the empty sequence. Given $w = w_1w_2 \in \Sigma^*$, we call w_1 a *prefix* of w . For a language L , its *prefix closure* $\bar{L}$ is defined as $\bar{L} = \{w_1 \in \Sigma^* \mid \exists w_2 \in \Sigma^* w_1w_2 \in L\}$. If $L = \bar{L}$, we call L *prefix closed*. The concatenation LL' of two languages is defined as $LL' = \{w_1w_2 \in \Sigma^* \mid w_1 \in L \wedge w_2 \in L'\}$.

We partition the set of events Σ into *controllable* events Σ_c that can be influenced by the supervisor and *uncontrollable* events Σ_u that the supervisor cannot affect. Traditionally (e.g., in [18]), a supervisor for plant $P \subseteq \Sigma^*$ is a function $f: P \rightarrow \Gamma$ such that $f(w) \in \Gamma$ prescribes the set of allowed events after w , which includes all uncontrollable events. Note that it is assumed that the event sets of the supervisor and the plant are the same. In this paper, the same assumption is made. Formally, let $\Gamma \subseteq 2^\Sigma$ be a set of control inputs, with $\Sigma_u \subseteq \gamma$ for all $\gamma \in \Gamma$. The supervised plant P_f is defined inductively such that $\varepsilon \in P_f$, and $wa \in P_f$ if $w \in P_f$, $a \in f(w)$ and $wa \in P$.

In practice, in particular when using tools, a finite description of languages is desired. Therefore, languages for the plant and the supervisor are often described using (finite) automata. This restricts the plant and the supervisor to regular languages. In some tools, such as Supremica [1] and CIF [5, 12], even extended finite automata (i.e., automata with (finite domain) state variables) are used for representing plant and supervisor [23]. In the remainder of this paper, we focus on plants and supervisors that are represented using automata.

In the context of supervisory control, an automaton is sometimes equipped with a finite set of marked states. These primarily play a role when considering nonblockingness. Since we are interested in studying controllability, and marked states do not play a role for that, we omit marked states from all definitions, even though some definitions of controllability in the literature take marked states into account.

Definition 1. An *automaton* G is a tuple $(Q_G, \Sigma, \rightarrow_G, q_G^0)$ where

- Q_G is a finite set of states;

- Σ is a finite set of events, partitioned into controllable events Σ_c and uncontrollable events Σ_u ;
- $\rightarrow_G \subseteq Q_G \times \Sigma \times Q_G$ is the transition relation;
- $q_G^0 \in Q_G$ is the initial state.

Note that the elements of the tuple are subscripted with the name of the automaton to easily identify which automaton is referred to. In case there can be no confusion these subscripts may be omitted. In the remainder of this paper, when clear from the context, we leave the sets of controllable (Σ_c) and uncontrollable events (Σ_u) implicit.

We typically use a, a_1, a_2 , etc. to denote arbitrary events, c, c_1, c_2 , etc. to denote arbitrary *control-*lable events, u, u_1, u_2 , etc. to denote arbitrary *uncontrollable* events, and w, w_1, w_2 , etc. for sequences of events. For $q, q' \in Q$ and $a \in \Sigma$, we typically write $q \xrightarrow{a} q'$ instead of $(q, a, q') \in \rightarrow$. We generalize this such that $q \xrightarrow{w} q'$ denotes the fact that there is a sequence of states connected by the events from the word $w \in \Sigma^*$ starting from state q and ending in state q' . We write $q \xrightarrow{a}$ (resp. $q \xrightarrow{w}$) to denote that there exists a $q' \in Q$ s.t. $q \xrightarrow{a} q'$ (resp. $q \xrightarrow{w} q'$). $\text{Reach}(G) = \{q \in Q \mid \exists_{w \in \Sigma^*} q_0 \xrightarrow{w} q\}$ denotes the set of reachable states of an automaton. Given a state $q \in Q$ and $a \in \Sigma$, we write $\Delta(q, a) = \{q' \in Q \mid q \xrightarrow{a} q'\}$.

If for all $q, q', q'' \in Q$ and $a \in \Sigma$, $q \xrightarrow{a} q'$ and $q \xrightarrow{a} q''$ implies $q' = q''$, we call the automaton *deterministic*. If G is deterministic and $q \xrightarrow{a}$, then $\Delta(q, a)$ is a singleton, and with slight abuse of notation, we write $\Delta(q, a)$ also for the single state that it contains.

Every automaton generates a discrete-event system in the traditional sense, i.e., as a prefix-closed language over alphabet Σ .

Definition 2. Let $G = (Q, \Sigma, \rightarrow, q^0)$ be an automaton. For state $q \in Q$, the *language* $L(q)$ generated by q is $L(q) = \{w \in \Sigma^* \mid q \xrightarrow{w}\}$. The *language* $L(G)$ generated by G is $L(G) = L(q^0)$.

Note that $L(G)$ is prefix closed, for all automata G . We typically simply write $L(G)$ instead of $\overline{L(G)}$.

In this paper, automata are depicted graphically whenever appropriate (see Fig. 1 for an example). States are denoted by circles containing the name of the state. The initial state is indicated by a dangling incoming arrow. Solid (resp. dashed) arrows indicate transitions labelled by controllable (resp. uncontrollable) events.

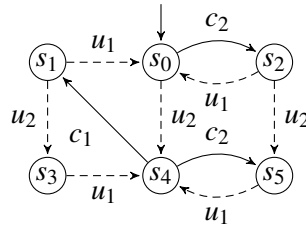


Figure 1: A graphical representation of an automaton.

The supervised plant is obtained as the synchronous product of plant and supervisor as defined below.

Definition 3. The *synchronous product* of automata $G = (Q_G, \Sigma, \rightarrow_G, q_G^0)$ and $H = (Q_H, \Sigma, \rightarrow_H, q_H^0)$, is the automaton $G \parallel H = (Q_{G \parallel H}, \Sigma, \rightarrow_{G \parallel H}, q_{G \parallel H}^0)$, where

- $Q_{G \parallel H} = Q_G \times Q_H$;
- $\rightarrow_{G \parallel H} = \{((q_1, q_2), a, (q'_1, q'_2)) \mid q_1 \xrightarrow{a}_G q'_1, q_2 \xrightarrow{a}_H q'_2\}$;
- $q_{G \parallel H}^0 = (q_G^0, q_H^0)$.

Sometimes we may assume that a supervisor S for a plant P is a subautomaton of that plant as defined below.

Definition 4 (Subautomaton). Automaton $G = (Q_G, \Sigma, \rightarrow_G, q_G^0)$ is a *subautomaton* of automaton $H = (Q_H, \Sigma, \rightarrow_H, q_H^0)$, denoted $G \subseteq H$, iff $Q_G \subseteq Q_H$, $\rightarrow_G \subseteq \rightarrow_H$, and $q_G^0 = q_H^0$.

4 Definitions of Controllability in the Literature

In this section, we define the controllability notions mentioned in Table 1. We present controllability notions as a binary relation between automata.

4.1 Language Controllability

In the literature, a notion of controllability is given (on a specification language and a plant language) and a controllability theorem is provided that states that controllability of that specification is a sufficient condition for existence of a supervisor for realizing that specification [8, 18]. In other work, controllability is also defined directly between a supervisor and a plant. To present all controllability notions in a single setting, we here define all notions of controllability between a supervisor and a plant.

When restricted to the languages of automata, this results in the following definition of controllability. This formulation is based on the observation that languages generated by automata are prefix closed. We refer to this notion as language controllability from now on to distinguish it from other notions of controllability defined later.

Definition 5 (Language controllability). Supervisor language $L(S) \subseteq \Sigma^*$ is language controllable w.r.t. plant language $L(P) \subseteq \Sigma^*$ if $L(S)\Sigma_u \cap L(P) \subseteq L(S)$. We write $L(S) \mathbf{LC} L(P)$ iff $L(S)$ is language controllable w.r.t. $L(P)$.

Intuitively, if after a string w in $L(S)$, an uncontrollable event u is possible in $L(P)$, then wu is also in the prefix closure of $L(S)$.

Lemma 1. Let S and P be automata over the same event set. Then $L(S) \mathbf{LC} L(P)$ iff $L(S) \parallel P \mathbf{LC} L(P)$.

Proof. Trivial. □

The other controllability notions we consider in this paper are defined as binary relations on the states of automata. Generally we say that automata are related iff their initial states are. For binary relations $R \subseteq Q_S \times Q_P$, we typically use infix notation $q R q'$ instead of $(q, q') \in R$.

To facilitate easier comparison between these notions and language controllability, we first reformulate language controllability on the states of automata.

Definition 6 (Automata controllability). Consider nondeterministic supervisor $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and nondeterministic plant $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$. A state $q_S \in Q_S$ is automata controllable w.r.t. state $q_P \in Q_P$, denoted $q_S \mathbf{AC} q_P$, iff for all $w \in \Sigma^*$, $u \in \Sigma_u$ if $q_P \xrightarrow{wu}_P$ and $q_S \xrightarrow{w}_S$, then also $q_S \xrightarrow{wu}_S$. Supervisor S is automata controllable w.r.t. P , denoted $S \mathbf{AC} P$ iff $q_S^0 \mathbf{AC} q_P^0$.

The notion of automata controllability and language controllability agree as expressed by the following lemma.

Lemma 2 (AC=LC). Let S and P be nondeterministic automata. Then $S \mathbf{AC} P$ iff $L(S) \mathbf{LC} L(P)$.

Proof. Let S and P be nondeterministic automata. We prove both directions separately.

- $\Rightarrow$ Suppose that $S \mathbf{AC} P$. We prove for all $w \in \Sigma^*$, $u \in \Sigma_u$, if $wu \in L(S)\Sigma_u \cap L(P)$, then $wu \in L(S)$. Pick arbitrary $wu \in L(S)\Sigma_u \cap L(P)$. Then, $wu \in L(P)$, so $q_P^0 \xrightarrow{wu}_P$, and $w \in L(S)$, so $q_S^0 \xrightarrow{w}_S$. As $S \mathbf{AC} P$, $q_S^0 \xrightarrow{wu}_S$, so $wu \in L(S)$, hence $L(S) \mathbf{LC} L(P)$.
- $\Leftarrow$ Suppose that $L(S) \mathbf{LC} L(P)$. We prove for all $w \in \Sigma^*$, $u \in \Sigma_u$, if $q_P^0 \xrightarrow{wu}_P$ and $q_S^0 \xrightarrow{w}_S$ then $q_S^0 \xrightarrow{wu}_S$. Pick arbitrary $w \in \Sigma^*$, $u \in \Sigma_u$ such that $q_P^0 \xrightarrow{wu}_P$ and $q_S^0 \xrightarrow{w}_S$, and note that $wu \in L(P)$ and $w \in L(S)$, so $wu \in L(S)\Sigma_u \cap L(P)$. As $L(S) \mathbf{LC} L(P)$, $wu \in L(S)$, so $q_S^0 \xrightarrow{wu}_S$. Hence $S \mathbf{AC} P$. $\square$

4.2 Control Relation

In [22], Rutten defines a *control relation* (**CR**) between two deterministic automata. This definition uses the fact that if an event is enabled in a certain state, then the target state is uniquely defined (due to the fact that the automaton is deterministic).

Definition 7 (Control relation). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be deterministic automata with uncontrollable event set Σ_u . A relation $R \subseteq Q_S \times Q_P$ is a *control relation* if, for all $q_S \in Q_S$ and $q_P \in Q_P$, $q_S R q_P$ implies

- for all $a \in \Sigma$ if $q_S \xrightarrow{a}_S$ and $q_P \xrightarrow{a}_P$, then $\Delta(q_S, a) R \Delta(q_P, a)$;
- for all $u \in \Sigma_u$ if $q_P \xrightarrow{u}_P$, then $q_S \xrightarrow{u}_S$ and $\Delta(q_S, u) R \Delta(q_P, u)$.

We write $q \mathbf{CR} q'$ iff there exists a control relation R such that $q R q'$; and write $S \mathbf{CR} P$ if $q_S^0 \mathbf{CR} q_P^0$.

The above definition does not require that any event enabled in a supervisor state must be enabled in a corresponding plant state. This suggests that this definition has been proposed with a supervised plant $S \parallel P$ in mind instead of a standalone supervisor S . In such a case, the supervised plant $S \parallel P$ would of course not enable the a event. We consider controllability of supervised plants in more detail in Section 6.

4.3 State Controllability

A notion of *state controllability* is defined in [24]. We here consolidate this notion in a single definition.

Definition 8 (State controllability). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ be a deterministic automaton and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be a (potentially nondeterministic) automaton. Then S is state controllable w.r.t. P , denoted $S \mathbf{SC} P$, iff

- there exists an embedding $f: Q_S \rightarrow Q_P$ such that
 - $f(q_S^0) = q_P^0$;
 - for all $q, q' \in Q_S$ and $a \in \Sigma$, if $q \xrightarrow{a}_S q'$ then $f(q) \xrightarrow{a}_P f(q')$;
- for all $q \in Q_S$, $F_S(q) \cap E_P(f(q)) \subseteq \Sigma_c$.

where $E_i(q) = \{a \in \Sigma \mid q' \in Q_i \wedge q \xrightarrow{a}_i q'\}$ and $F_i(q) = \Sigma \setminus E_i(q)$.¹

¹The definition in [24] furthermore considers marked states, and requires that $f(Q_S^m) \subseteq Q_P^m$.

4.4 Partial Bisimilarity

Rutten introduced partial bisimilarity in a coalgebraic setting to capture controllability of deterministic systems [22].

Definition 9 (Partial bisimulation). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be deterministic automata. Relation $R \subseteq Q_S \times Q_P$ is a *partial bisimulation* if, for all $q_S \in Q_S$ and $q_P \in Q_P$, $q_S R q_P$ implies

- for all $a \in \Sigma$ if $q_S \xrightarrow{a}_S$ then $q_P \xrightarrow{a}_P$ and $\Delta(q_S, a) R \Delta(q_P, a)$;
- for all $u \in \Sigma_u$ if $q_P \xrightarrow{u}_P$ then $q_S \xrightarrow{u}_S$ and $\Delta(q_S, u) R \Delta(q_P, u)$.

We write $q_S \mathbf{PB} q_P$ if there exists a partial bisimulation relation R such that $q_S R q_P$. We also call relation $\mathbf{PB}$ *partial bisimilarity*.²

Intuitively, the main difference between control relations and partial bisimulation is that a control relation allows the supervisor to execute controllable events that are not enabled in a related plant state. In partial bisimilarity this is not allowed.

4.5 Nondeterministic Partial Bisimilarity

Partial bisimilarity was generalized to nondeterministic automata [3, 17]. In [15] partial bisimilarity is used to capture controllability in the context of synthesis of maximally permissive supervised plants for nondeterministic plants and (a subset of) modal logic specifications. The generalization to nondeterministic automata is as follows.

Definition 10. (Nondeterministic partial bisimulation) Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata. Relation $R \subseteq Q_S \times Q_P$ is a *nondeterministic partial bisimulation* if, for all $q_S \in Q_S$ and $q_P \in Q_P$, $q_S R q_P$ implies

- for all $a \in \Sigma, q'_S \in Q_S$ such that $q_S \xrightarrow{a}_S q'_S$, there exists $q'_P \in Q_P$ such that $q_P \xrightarrow{a}_P q'_P$ and $q'_S R q'_P$;
- for all $u \in \Sigma_u, q'_P \in Q_P$ such that $q_P \xrightarrow{u}_P q'_P$, there exists $q'_S \in Q_S$ such that $q_S \xrightarrow{u}_S q'_S$ and $q'_S R q'_P$.

We write $q_S \mathbf{PB}^n q_P$ if there exists a nondeterministic partial bisimulation relation R such that $q_S R q_P$. This relation is called *nondeterministic partial bisimilarity*. We say that $S \mathbf{PB}^n P$ iff $q_S^0 \mathbf{PB}^n q_P^0$.²

4.6 Controllability for nondeterministic automata

Another lifting of controllability to the nondeterministic automata setting is, e.g., used by [10] and [11]. We abbreviate this notion as **FM**-controllability.

Definition 11 (FM-Controllability). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata. A state $q_S \in Q_S$ is **FM**-controllable w.r.t. state $q_P \in Q_P$, denoted $q_S \mathbf{FM} q_P$, iff for all $w \in \Sigma^*$, $u \in \Sigma_u$ and $q \in Q_S$, if $q_P \xrightarrow{wu}_P$ and $q_S \xrightarrow{w}_S q$, then also $q \xrightarrow{u}_S$. Supervisor S is **FM**-controllable w.r.t. P , denoted $S \mathbf{FM} P$, iff $q_S^0 \mathbf{FM} q_P^0$.

Note that this definition is slightly stronger than automata controllability. That is, **FM**-controllability requires that from the specific state q such that $q_S \xrightarrow{w}_S q$, it is possible to take the u -labelled transition, whereas this is not required by automata controllability.

²The definitions in [3, 17, 22] take marked states into account and additionally require that $q_S R q_P$ implies $q_S \in Q_S^m$ iff $q_P \in Q_P^m$.

4.7 Σ_u -Admissibility

Kushi and Takai [16], define Σ_u -admissibility. Essentially a supervisor is Σ_u -admissible if reachable states in the supervised plant do not disable uncontrollable actions.

Definition 12 (Σ_u -Admissibility). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata. Then S is Σ_u -admissible w.r.t. P if for all $(q_S, q_P) \in \text{Reach}(S \parallel P)$ and $u \in \Sigma_u$, if $q_P \xrightarrow{u}_P$ then $(q_S, q_P) \xrightarrow{u}_{S \parallel P}$. For the sake of brevity, we write $S \mathbf{KT} P$ iff S is Σ_u -admissible w.r.t. P .

4.8 Nondeterministic State Controllability

For a setting where plant, specification and supervisor are represented by nondeterministic automata, in [26], state controllability is defined of a specification w.r.t. a plant. This definition is restricted to cases where the language of the specification is contained in the language of the plant. Also, the definition is partially formulated using languages instead of automata.

Definition 13 (State controllability). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata such that $L(S) \subseteq L(P)$. Then, S is state-controllable w.r.t. P iff for all $w \in L(S)$ and $u \in \Sigma_u$ such that $wu \in L(P)$, for all $q \in Q_S$, if $q_S^0 \xrightarrow{w}_S q$, then $q \xrightarrow{u}_S$. For the sake of brevity, we write $S \mathbf{SC}^n P$ iff S is state-controllable w.r.t. P .

Note that the restriction $L(S) \subseteq L(P)$ in the definition only has limited effect. If we drop the restriction, for $w \in L(S) \setminus L(P)$, the definition does not impose any restrictions since $w \notin L(P)$ implies $wu \notin L(P)$ for all $u \in \Sigma_u$. Kushi and Takai [16] show that S is nondeterministic state controllable w.r.t. P iff S is Σ_u -admissible w.r.t. P .

4.9 Controllability for nondeterministic subautomata

If the supervisor is a subautomaton of the plant and the supervisor can disable individual transitions, [10] and [11] use the following variation of $\mathbf{FM}$ -controllability, that we abbreviate $\mathbf{FM}_{\subseteq}$ -controllability.

Definition 14 ($\mathbf{FM}_{\subseteq}$ -Controllability). Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata such that $S \subseteq P$. A state $q_S \in Q_S$ is $\mathbf{FM}_{\subseteq}$ -controllable w.r.t. state $q_P \in Q_P$, denoted $q_S \mathbf{FM}_{\subseteq} q_P$, iff for all $w \in \Sigma^*$, $u \in \Sigma_u$ and q, q' if $q_P \xrightarrow{w}_P q \xrightarrow{u}_P q'$ and $q_S \xrightarrow{w}_S q$, then also $q \xrightarrow{u}_S q'$. Supervisor S is $\mathbf{FM}_{\subseteq}$ -controllable w.r.t. P , denoted $S \mathbf{FM}_{\subseteq} P$, iff $q_S^0 \mathbf{FM}_{\subseteq} q_P^0$.

As mentioned by [10], in case the supervisor is a subautomaton of a *deterministic* plant, the notions of $\mathbf{FM}$ -controllability and $\mathbf{FM}_{\subseteq}$ -controllability coincide.

5 Comparing Notions of Controllability

We next compare the notions of controllability as they have been defined in the literature, for the contexts to which they can be applied. When C_1 and C_2 are controllability notions, we write $C_1 \subseteq C_2$ if C_1 is stronger than C_2 , $C_1 = C_2$ if C_1 and C_2 are equivalent, i.e., $C_1 \subseteq C_2$ and $C_2 \subseteq C_1$, $C_1 \subsetneq C_2$ if $C_1 \subseteq C_2$ and not $C_1 = C_2$. We write $C_1 \# C_2$ if C_1 and C_2 are incomparable, that is, neither $C_1 \subseteq C_2$ nor $C_2 \subseteq C_1$.

In the remainder of this section, we first compare the notions of controllability that do not require the (language of the) supervisor to be a subautomaton of the plant. The latter cases are compared separately. The notions are compared for each of the contexts for which they are defined.

5.1 Nondeterministic Supervisors and Plants

In this section, comparison of the previously defined notions of controllability is provided restricted to those notions that are defined for nondeterministic supervisors and plants. These are language (**LC**) and automata controllability (**AC**), nondeterministic partial bisimilarity (**PBⁿ**), **FM**-controllability (**FM**) and Σ_u -admissibility (**KT**).

Lemma 3 (FM=KT). *For nondeterministic automata S and P : $S \text{ FM } P$ iff $S \text{ KT } P$.*

Proof. Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata. We prove both directions separately.

$\Rightarrow$ Suppose that $S \text{ FM } P$. Let $(q_S, q_P) \in \text{Reach}(S \parallel P)$ and let $u \in \Sigma_u$. Assume that $q_P \xrightarrow{u} p$. We need to show that $(q_S, q_P) \xrightarrow{u}_{S \parallel P}$. Since $(q_S, q_P) \in \text{Reach}(S \parallel P)$, we have the existence of $w \in \Sigma^*$ such that $(q_S^0, q_P^0) \xrightarrow{w}_{S \parallel P} (q_S, q_P)$. Therefore, as S and P have the same event set and by definition of $\parallel$, we obtain $q_S^0 \xrightarrow{w}_S q_S$ and $q_P^0 \xrightarrow{w}_P q_P$. Combining $q_P^0 \xrightarrow{w}_P q_P$ and $q_P \xrightarrow{u} p$ gives $q_P^0 \xrightarrow{wu}_P p$. Since S and P are **FM**-controllable, combined with $q_P^0 \xrightarrow{wu}_P p$ and $q_S^0 \xrightarrow{w}_S q_S$, it follows that $q_S \xrightarrow{u}_S$. Thus we have both $q_S \xrightarrow{u}_S$ and $q_P \xrightarrow{u}_P$, and thus $(q_S, q_P) \xrightarrow{u}_{S \parallel P}$ as well, which was to be proven.

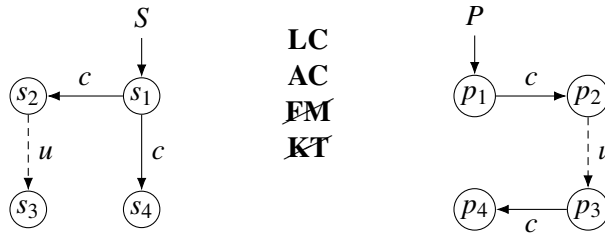
$\Leftarrow$ Suppose that $S \text{ KT } P$. Let $w \in \Sigma^*$, $u \in \Sigma_u$ and $q_S \in Q_S$. Assume that $q_P^0 \xrightarrow{wu}_P$ and $q_S^0 \xrightarrow{w}_S q_S$. We need to show that $q_S \xrightarrow{u}_S$. From $q_P^0 \xrightarrow{wu}_P$ we have the existence of $q_P \in Q_P$ such that $q_P^0 \xrightarrow{w}_P q_P$ and $q_P \xrightarrow{u}_P$. Since S and P have the same event set, and by definition of $\parallel$ it follows that $(q_S^0, q_P^0) \xrightarrow{w}_{S \parallel P} (q_S, q_P)$. Since $(q_S, q_P) \in \text{Reach}(S \parallel P)$ and $q_P \xrightarrow{u}_P$ and because S is Σ_u -admissible w.r.t. P , we have $(q_S, q_P) \xrightarrow{u}_{S \parallel P}$. Thus, also $q_S \xrightarrow{u}_S$, which was to be shown. $\square$

Lemma 4 (FM $\subseteq$ AC). *For nondeterministic automata S and P : If $S \text{ FM } P$, then $S \text{ AC } P$.*

Proof. Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata. Assume $S \text{ FM } P$. Let $w \in \Sigma^*$, $u \in \Sigma_u$. Assume $q_P^0 \xrightarrow{wu}_P$ and $q_S^0 \xrightarrow{w}_S$. We need to prove that $q_S^0 \xrightarrow{wu}_S$. From $q_S^0 \xrightarrow{w}_S$ we have $q_S^0 \xrightarrow{w}_S q_S$ for some $q_S \in Q_S$. As $S \text{ FM } P$, we have $q_S \xrightarrow{u}_S$, and consequently $q_S^0 \xrightarrow{wu}_S$. $\square$

The following example shows that automata controllability is not stronger than **FM**-controllability and Σ_u -admissibility.

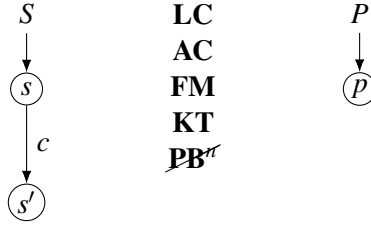
Example 1. Consider the following supervisor S and plant P .



We have $S \text{ LC } P$ and $S \text{ AC } P$. However, $S \not\text{ FM } P$ as $p_1 \xrightarrow{cu}$, and $s_1 \xrightarrow{c} s_4$, but $s_4 \not\xrightarrow{u}$, and $S \not\text{ KT } P$ as $\text{FM}=\text{KT}$. $\blacksquare$

The following example shows that automata controllability, **FM**-controllability and Σ_u -admissibility are not stronger than nondeterministic partial bisimilarity.

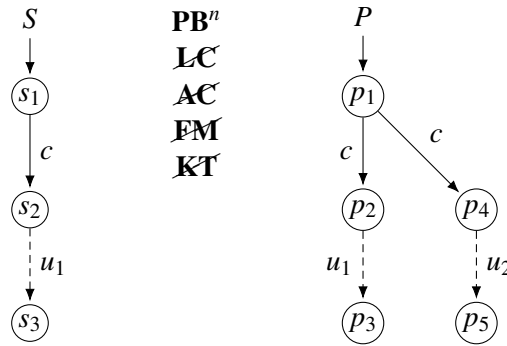
Example 2. Consider the following supervisor S and plant P , both with event set $\Sigma = \{c\}$.



We have $S \mathbf{AC} P$, $S \mathbf{FM} P$, and $S \mathbf{KT} P$, but not $S \mathbf{PB}^n P$ because no partial bisimulation relation between s and p exists as p cannot mimic the c -transition. So, $\mathbf{AC} \not\subseteq \mathbf{PB}^n$, $\mathbf{FM} \not\subseteq \mathbf{PB}^n$, and $\mathbf{KT} \not\subseteq \mathbf{PB}^n$. ■

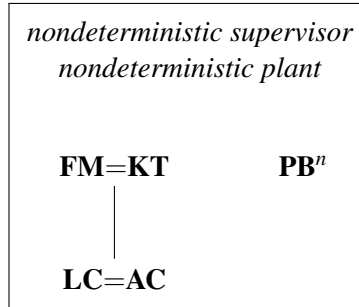
The next example shows that nondeterministic partial bisimilarity is not stronger than automata controllability, **FM**-controllability and Σ_u -admissibility.

Example 3. Consider the following supervisor S and plant P , with $\Sigma_u = \{u_1, u_2\}$.



We have that $R = \{(s_1, p_1), (s_2, p_2), (s_3, p_3)\}$ is a partial bisimulation relation, so $S \mathbf{PB}^n P$. However, $S \mathbf{AC} P$, since $p_1 \xrightarrow{cu_2} p$, $s_1 \xrightarrow{c} s$, but $s_1 \not\xrightarrow{cu_2} s$. The argumentation for $S \mathbf{FM} P$ and $S \mathbf{KT} P$ are similar. ■

Theorem 1. In case the supervisor and plant are nondeterministic, we have the following identifications and relations.



Proof. That $\mathbf{LC}=\mathbf{AC}$ follows from Lemma 2; $\mathbf{FM}=\mathbf{KT}$ follows from Lemma 3; $\mathbf{FM} \subseteq \mathbf{AC}$ and $\mathbf{KT} \subseteq \mathbf{AC}$ follow from Lemma 4 and $\mathbf{FM}=\mathbf{KT}$; $\mathbf{AC} \not\subseteq \mathbf{FM}$ follows from Example 1; and $\mathbf{PB}^n \# \mathbf{FM}$, $\mathbf{PB}^n \# \mathbf{KT}$ and $\mathbf{PB}^n \# \mathbf{AC}$ follow from Examples 2 and 3. □

5.2 Deterministic Supervisors and Nondeterministic Plants

In addition to the notions in the previous subsection, for deterministic supervisors and nondeterministic plants also state controllability is defined. We add this to the comparison, and reconsider the comparisons between the previous notions.

Since this context is more specific, all implications found in the previous subsection remain valid, but additional ones emerge. Since we have already established that $\mathbf{FM}=\mathbf{KT}$, we forego the discussion of Σ_u -admissibility ($\mathbf{KT}$) in the lemmas in this section, and only consider $\mathbf{FM}$ -controllability.

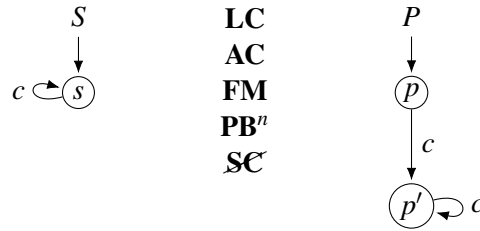
In this context, automata controllability is stronger than $\mathbf{FM}$ -controllability.

Lemma 5 ($\mathbf{AC} \subseteq \mathbf{FM}$). *Let S be a deterministic automaton and P a (possibly nondeterministic) automaton. If $S \mathbf{AC} P$, then $S \mathbf{FM} P$.*

Proof. Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ be a deterministic automaton and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ a nondeterministic automaton. Assume that $S \mathbf{AC} P$. We show that $S \mathbf{FM} P$. Fix $w \in \Sigma^*$, $u \in \Sigma_u$ and $q \in Q_S$ such that $q_P^0 \xrightarrow{wu}_P$ and $q_S^0 \xrightarrow{w}_S q$. Since $S \mathbf{AC} P$, $q_S^0 \xrightarrow{wu}_S$; since S is deterministic, $q = \Delta(w)$ is unique, so $q \xrightarrow{u}_S$. Therefore, $S \mathbf{FM} P$. $\square$

The following example shows that automata controllability, $\mathbf{FM}$ -controllability and nondeterministic partial bisimilarity are not stronger than state controllability.

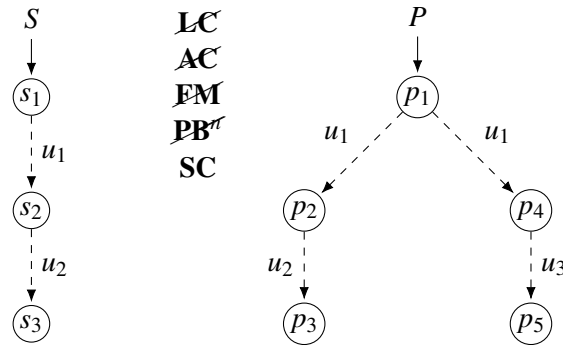
Example 4. Consider the following deterministic supervisor S and plant P .



Note that $R = \{(s, p), (s, p')\}$ is a nondeterministic partial bisimulation relation. However, there exists no embedding f of S to P witnessing $S \mathbf{SC} P$, so $\mathbf{PB}^n \not\subseteq \mathbf{SC}$. Since the plant and supervisor do not have any uncontrollable events, $S \mathbf{LC} P$, $S \mathbf{AC} P$ and $S \mathbf{FM} P$ trivially hold. $\blacksquare$

Conversely, state controllability is not stronger than automata controllability, $\mathbf{FM}$ -controllability and nondeterministic partial bisimilarity, as shown by the following example.

Example 5. Consider the following deterministic supervisor S and nondeterministic supervisor P , with $\Sigma_u = \{u_1, u_2, u_3\}$.



We have that $S \mathbf{SC} P$, witnessed by $f(s_1) = p_1, f(s_2) = p_2$ and $f(s_3) = p_3$. In particular, no uncontrollable events are forbidden in p_1 , even though the transition to p_4 is removed. However, $S \mathbf{AC} P$, since $p_1 \xrightarrow{u_1 u_3}_P$, $s_1 \xrightarrow{u_1}_S$, but $s_1 \not\xrightarrow{u_1 u_3}_S$; as $\mathbf{LC}=\mathbf{AC}$, also $S \mathbf{LC} P$. The argumentation for $S \mathbf{FM} P$ is similar. Also, there is no partial bisimulation relation relating s_2 in the supervisor and p_4 in the plant, since s_2 cannot mimic the transition to p_5 , however, in any partial bisimulation relation, s_2 and p_4 must be related, so $S \mathbf{PB}^\pi P$. $\blacksquare$

Theorem 2. *In case the supervisor is deterministic and the plant is nondeterministic, we have the following identifications and relations.*

deterministic supervisor nondeterministic plant		
LC=AC=FM=KT	PBⁿ	SC

Proof. That **LC=AC** follows from Lemma 2; **FM=KT** follows from Lemma 3; we do not explicitly consider **KT** explicitly in the remaining arguments. That **AC=FM** follows from Lemmas 4 and 5; **PBⁿ#AC** and **PBⁿ#FM** follow from Examples 2 and 3, which remain valid since S is deterministic in both examples. That **SC#AC**, **SC#FM**, and **SC#PBⁿ** follow from Examples 4 and 5. $\square$

5.3 Deterministic Supervisors and Plants

When we consider both supervisor and plant to be deterministic, two additional notions need to be compared. These are control relations (**CR**) and partial bisimilarity (**PB**).

Rutten [22] established that for deterministic supervisors and plants, language controllability and control relations coincide.

Lemma 6 (LC=CR). *Let S and P be deterministic automata. Then $S \text{ LC } P$ iff $S \text{ CR } P$.*

That partial bisimilarity and nondeterministic partial bisimilarity coincide in this setting follows immediately from the definitions.

Lemma 7 (PB=PBⁿ). *Let S and P be deterministic automata. Then $S \text{ PB } P$ iff $S \text{ PB}^n P$.*

Rutten also showed that on deterministic automata, partial bisimilarity is a control relation [22].

Lemma 8 (PB $\subseteq$ CR). *Let S and P be deterministic automata. If $S \text{ PB } P$, then $S \text{ CR } P$.*

Note that the converse does not hold.

Example 6. Recall supervisor S and plant P from Example 2. Both are deterministic. As $S \text{ AC } P$ and **AC=CR**, $S \text{ CR } P$; from the example also $S \text{ PB}^n P$, so combined with **PB=PBⁿ** this gives **CR $\not\subseteq$ PB**. $\blacksquare$

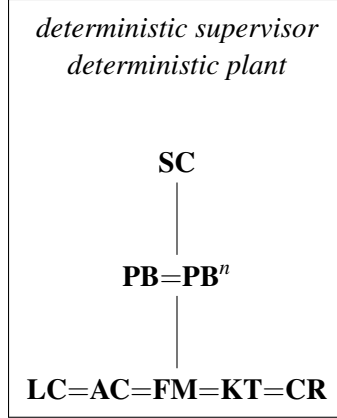
Finally we establish that state controllability is stronger than partial bisimilarity.

Lemma 9 (SC $\subseteq$ PB). *Let S and P be deterministic automata. If $S \text{ SC } P$, then $S \text{ PB } P$.*

Proof. Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be deterministic automata, and suppose $S \text{ SC } P$. Let $f: Q_S \rightarrow Q_P$ be the embedding that witnesses the state controllability and define the relation $R = \{(s, f(s)) \mid s \in Q_S\}$. We show that R is a partial bisimulation relation. Pick arbitrary q_S, q_P such that $q_S R q_P$. By definition of R , $q_P = f(q_S)$. We check the conditions of partial bisimilarity.

- Let $a \in \Sigma$, $q'_S \in Q_S$ and $q_S \xrightarrow{a}_S q'_S$. According to the definition of state controllability, $q_P = f(q_S) \xrightarrow{a}_P f(q'_S)$, and by definition of R , $q'_S R f(q'_S)$.
- Let $u \in \Sigma_u$, $q'_P \in Q_P$ and $q_P \xrightarrow{u}_P q'_P$. By definition of state controllability, there exists q'_S such that $q_S \xrightarrow{u}_S q'_S$. By the definition of state controllability, $q_P = f(q_S) \xrightarrow{u}_P f(q'_S)$. Since P is deterministic $f(q_S) = q'_P$, and since $q'_S R f(q'_S) = q'_P$, the result follows. $\square$

Theorem 3. *In case the supervisor and plant are deterministic, we have the following identifications and relations.*



Proof. Theorem 2 provides us with **LC=AC=FM=KT**, also for deterministic supervisors and plants. Hence, we do not consider **AC**, **FM** and **KT** any further in the remaining proofs. That **LC=CR** follows from Lemma 6; and **PB=PBⁿ** follows from Lemma 7. For the remaining proofs we only need to consider one notion from each equivalence class. We only consider **SC**, **PB** and **CR**. That **PB** ⊆ **CR** follows from Lemma 8; **CR** ⊈ **PB** follows from Example 6; **SC** ⊆ **PB** follows from Lemma 9; and **PB** ⊈ **SC** follows from Example 4 since **PB=PBⁿ**. □

5.4 Subautomata

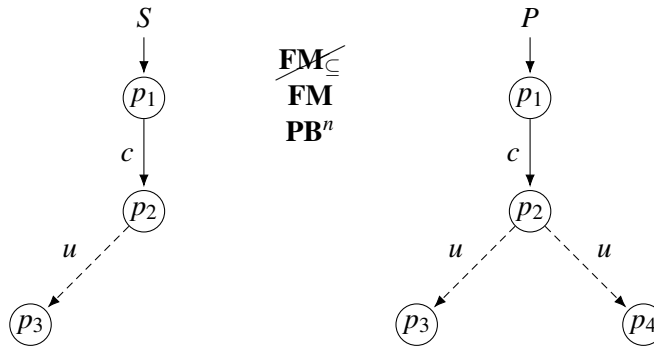
Many implementations deciding controllability use the premise that the supervisor is a subautomaton of the plant, that is $S \subseteq P$. In this case, additionally **FM**_⊆ and **SC**ⁿ are defined (as $S \subseteq P$ implies $L(S) \subseteq L(P)$). Both notions are defined for the general context of nondeterministic supervisor and plant. For the sake of brevity we restrict our exposition to that context.

The following relation between **FM**_⊆ and **FM** was established by Flordal and Malik [10, 11].

Lemma 10 (FM_⊆ ⊆ FM). *Let S and P be nondeterministic automata such that $S \subseteq P$. If $S \text{ FM}_{\subseteq} P$, then $S \text{ FM } P$.*

The converse does not hold. This is shown by the following example.

Example 7. Consider the supervisor S and plant P with $S \subseteq P$, and $\Sigma_u = \{u\}$ as depicted below.



That $S \text{ FM } P$ follows from the fact that for the only sequence of events in P that ends with an uncontrollable event, i.e., $p_1 \xrightarrow{cu} p$, and the only transition with c in S , i.e., $p_1 \xrightarrow{c} p_2$, we also have $p_2 \xrightarrow{u} p$. We cannot have $S \text{ FM}_{\subseteq} P$, since the sequence $p_1 \xrightarrow{c} p_2 \xrightarrow{u} p_4$ and the sequence $p_1 \xrightarrow{c} p_2$ are present but not $p_2 \xrightarrow{u} p_4$, as required by **FM**_⊆. Note that we also have $S \text{ PB}^n P$ as witnessed by the nondeterministic partial bisimulation $\{(p_1, p_1), (p_2, p_2), (p_3, p_3), (p_3, p_4)\}$. ■

Lemma 11. *Let S and P be nondeterministic automata such that $L(S) \subseteq L(P)$. Then $S \mathbf{SC}^n P$ iff $S \mathbf{FM} P$.*

Proof. Immediate from the definitions. As a side note, $\mathbf{SC}^n = \mathbf{KT}$ is already shown in [16]. Combined with $\mathbf{FM} = \mathbf{KT}$ this equality also proves the lemma. $\square$

Since $S \subseteq P$ implies $L(S) \subseteq L(P)$, we immediately have that Lemma 11 also holds for the context discussed in this section.

Lemma 12 ($\mathbf{FM}_{\subseteq} \subseteq \mathbf{PB}^n$). *Let S and P be nondeterministic automata such that $S \subseteq P$. If $S \mathbf{FM}_{\subseteq} P$ then $S \mathbf{PB}^n P$.*

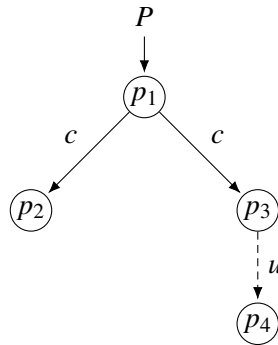
Proof. Let $S = (Q_S, \Sigma, \rightarrow_S, q_S^0)$ and $P = (Q_P, \Sigma, \rightarrow_P, q_P^0)$ be nondeterministic automata and suppose $S \mathbf{FM}_{\subseteq} P$. We define the relation $R = \{(q_S, q_P) \in Q_S \times Q_P \mid q_S \in \text{Reach}(S)\}$. We show that R is a nondeterministic partial bisimulation relation. Pick arbitrary q_S such that $q_S R q_P$.

- For the transfer condition from left to right, pick arbitrary $q'_S \in Q_S$, $a \in \Sigma$ such that $q_S \xrightarrow{a}_S q'_S$. As $S \subseteq P$, $q'_S \in Q_P$, and $q_S \xrightarrow{a}_P q'_S$, and by definition of R , $q'_S R q'_S$.
- For the transfer condition from right to left, pick arbitrary $q'_P \in Q_P$, $u \in \Sigma_u$ such that $q_S \xrightarrow{u}_P q'_P$. As q_S is reachable in S , and $S \subseteq P$, q_S is reachable in P . Let $w \in \Sigma^*$ be a witness such that $q_S^0 \xrightarrow{w}_S q_S$ and $q_S^0 \xrightarrow{w}_P q_S$. So, we have $q_S^0 \xrightarrow{w}_P q_S \xrightarrow{u}_P q'_P$ and $q_S^0 \xrightarrow{w}_S q_S$. As $S \mathbf{FM}_{\subseteq} P$ we then also have $q_S \xrightarrow{u}_S q'_P$, so $q'_P \in Q_S$, hence $q'_P R q'_P$.

So, R is a nondeterministic partial bisimulation relation, hence $S \mathbf{PB}^n P$. $\square$

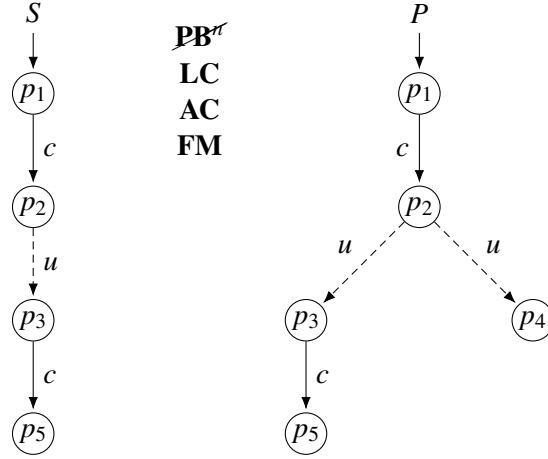
The following example shows that, even if $S \subseteq P$, automata controllability is not stronger than $\mathbf{FM}$ -controllability.

Example 8. Consider the plant P with $\Sigma_u = \{u\}$ as depicted below, and a supervisor $S = P$.



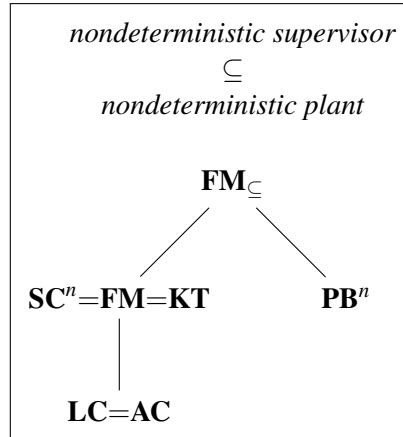
Note that trivially $S \subseteq P$, and that $S \mathbf{AC} P$. However, $S \not\mathbf{FM} P$, because $p_1 \xrightarrow{cu}_P$ and $p_1 \xrightarrow{c}_S p_2$, but not $p_2 \xrightarrow{u}_S$. $\blacksquare$

Example 9. Consider the supervisor S and plant P with $S \subseteq P$, and $\Sigma_u = \{u\}$ as depicted below.



$S \mathbf{FM} P$ follows from the fact that cu is the only sequence of events in P that ends in an uncontrollable event, i.e., $p_1 \xrightarrow{cu}_P$, in S , $p_1 \xrightarrow{c}_S p_2$ is the only c -transition from p_1 , and we also have $p_2 \xrightarrow{u}_S p_3$. $S \mathbf{AC} P$ follows from this and $\mathbf{FM} \subseteq \mathbf{AC}$. However, every nondeterministic partial bisimulation relation must relate p_2 in S and P , which means that the transition $p_2 \xrightarrow{u}_P p_4$ must be mimicked by some transition in S . Such transition does not exist. ■

Theorem 4. *In case the supervisor is a nondeterministic subautomaton of a nondeterministic plant, we have the following identifications and relations.*



Proof. Theorem 1 provides us with $\mathbf{LC} = \mathbf{AC}$, $\mathbf{FM} = \mathbf{KT}$ and $\mathbf{FM} \subseteq \mathbf{AC}$; that $\mathbf{SC}^n = \mathbf{FM}$ follows from Lemma 11. That $\mathbf{FM} \subsetneq \mathbf{FM}$ follows from Lemma 10 and Example 7; $\mathbf{FM} \subsetneq \mathbf{PB}^n$ follows from Lemma 12 and Example 7; $\mathbf{AC} \not\subseteq \mathbf{FM}$ follows from Example 8; $\mathbf{PB}^n \not\subseteq \mathbf{AC}$ follows from Example 3 provided that s_1 , s_2 and s_3 are renamed into p_1 , p_2 and p_3 , respectively, in the supervisor (for it to be a subautomaton of P); $\mathbf{AC} \not\subseteq \mathbf{PB}^n$ follows from Example 9; $\mathbf{PB}^n \not\subseteq \mathbf{FM}$ follows from $\mathbf{FM} \subseteq \mathbf{AC}$ and $\mathbf{PB}^n \not\subseteq \mathbf{AC}$. $\mathbf{FM} \not\subseteq \mathbf{PB}^n$ follows from Example 9. □

5.5 Reflection

In the previous subsections we have seen that, depending on the context, very different relationships between the notions emerge. Only $\mathbf{PB}^n$, $\mathbf{KT}$, $\mathbf{FM}$, and $\mathbf{AC}$ are defined for the most general context of nondeterministic plants and supervisors. Additionally, all these notions, except for $\mathbf{PB}^n$, are proper generalizations of language controllability according to Theorem 1.

Nondeterministic automata appear naturally, for example, when considering partial observability [9]. However, nondeterministic automata are also often used as a means to represent deterministic automata in a more convenient way. If that is the case, then one would expect that the controllability notion is invariant w.r.t. determinization, i.e., (a) $S \mathbf{X} P$ iff $Det(S) \mathbf{X} P$ and (b) $S \mathbf{X} P$ iff $S \mathbf{X} Det(P)$, where Det provides a language equivalent deterministic automaton (see [14] for example).

Automata controllability (**AC**) is invariant w.r.t. determinization, as stated in the following lemma.

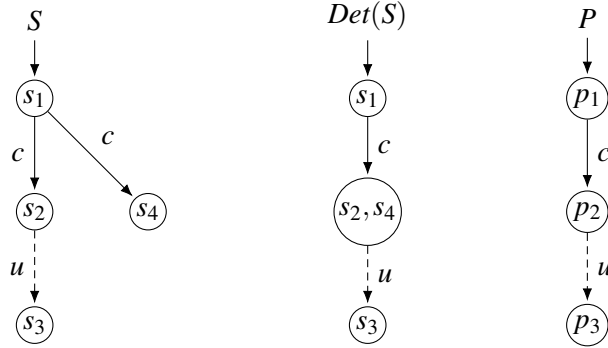
Lemma 13. *Let S and P be nondeterministic automata. Then*

- $S \mathbf{AC} P$ iff $Det(S) \mathbf{AC} P$;
- $S \mathbf{AC} P$ iff $S \mathbf{AC} Det(P)$.

Proof. This lemma follows immediately from the fact that for any automaton G and any $w \in \Sigma^*$, $q_G^0 \xrightarrow{w}_G$ iff $q_{Det(G)}^0 \xrightarrow{w}_{Det(G)}$. $\square$

If a controllability notion that generalizes language controllability is invariant under determinization, then we call it a trivial generalization to nondeterministic automata, and a non-trivial generalization otherwise. The following example shows that the other remaining notions under consideration, i.e., **FM** and **KT** are not invariant under determinization of the supervisor. This means that these are non-trivial generalizations to nondeterministic automata.

Example 10. Consider the following supervisor S and plant P , with $\Sigma_u = \{u\}$.



Note that $Det(S) \mathbf{FM} P$, but not $S \mathbf{FM} P$. Then, as $\mathbf{FM} = \mathbf{KT}$ the same conclusion can be drawn for **KT** in place of **FM**. $\blacksquare$

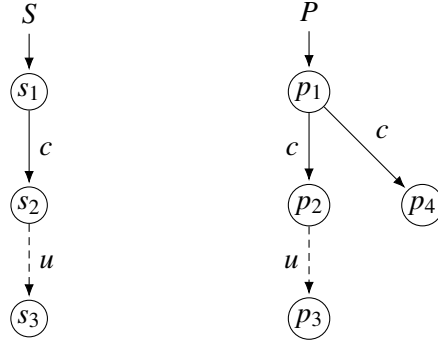
Concluding, assuming that we are most interested in a notion that is defined for the most general context, and that generalizes language controllability in a non-trivial way, we can work with each of the notions **FM**-controllability and Σ_u -admissibility. Note that these have been established to be equivalent in all contexts.

6 Controllability of Supervised Plant

In the previous section, the notion of controllability was defined on a supervisor and a plant. From a practical point of view, it is not so relevant to have controllability of the supervisor w.r.t. the plant, but it is necessary to have controllability of the supervised plant w.r.t. the plant.

In the language-based setting (see Lemma 1), it is always the case that $S \parallel P$ is controllable for P whenever S is controllable for P , since the supervised plant is achieved as $S \parallel P$. However, for controllability notions such as **FM** and **KT**, this is not the case, as detailed below.

Example 11. Consider the following supervisor S and plant P , with $\Sigma_u = \{u\}$.



Note that $S \mathbf{FM} P$, but not $(S \parallel P) \mathbf{FM} P$. We find $p_1 \xrightarrow{cu}_P$ and $(s_1, p_1) \xrightarrow{c}_{S \parallel P} (s_2, p_4)$, but $(s_2, p_4) \not\xrightarrow{u}_{S \parallel P}$. Note that since $\mathbf{FM} = \mathbf{KT}$, this implies that the same conclusions can be drawn for $\mathbf{KT}$ as well. ■

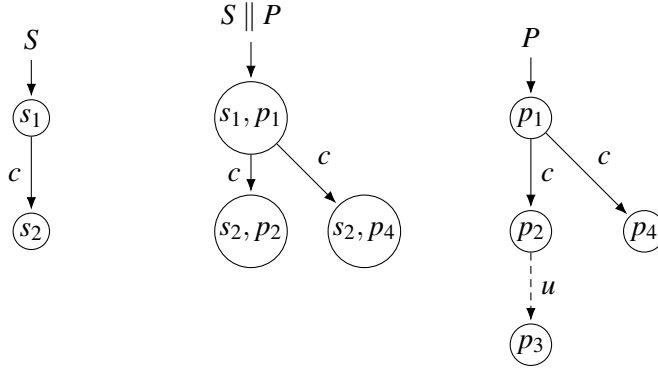
In the remainder of this section we will briefly reconsider all defined notions of controllability where we consider supervised plant $S \parallel P$ versus plant P instead of supervisor S versus P as we did before. With this interpretation, besides $\mathbf{AC}$, $\mathbf{FM}$, $\mathbf{KT}$ and $\mathbf{PB}^n$, also $\mathbf{SC}^n$ is defined in the most general context since $L(S \parallel P) \subseteq L(P)$ (as required for $\mathbf{SC}^n$).

For $\mathbf{AC}$, $\mathbf{KT}$ and $\mathbf{FM}$, the fact that they generalize language controllability for arbitrary supervisor implies that this is also the case for the supervised plant. So these three notions still generalize language controllability. Since $L(S \parallel P) \subseteq L(P)$, following Lemma 11, also $\mathbf{SC}^n$ generalizes language controllability. In contrast $\mathbf{PB}^n$ does not generalize language controllability as shown in the following example.

Example 12. Consider the following supervisor S and plant P . Note that $(S \parallel P) \mathbf{PB}^n P$ as witnessed by the partial bisimulation relation

$$\{((s_1, p_1), p_1), ((s_2, p_2), p_4), ((s_2, p_4), p_4)\}.$$

However, it is not the case that $(S \parallel P) \mathbf{AC} P$ because $p_1 \xrightarrow{cu}_P$, but not $(s_1, p_1) \xrightarrow{cu}_{S \parallel P}$.



Since $\mathbf{LC} = \mathbf{AC}$, also $(S \parallel P) \mathbf{LC} P$. ■

The fact that $S \parallel P = S = \text{Det}(S)$ in Example 11, shows that $\mathbf{FM}$ and $\mathbf{KT}$ are still non-trivial generalizations to the nondeterministic context. Also $\mathbf{SC}^n$ is a non-trivial generalization. In Example 11, again $S \parallel P = S$ and not $(S \parallel P) \mathbf{SC}^n P$, while $\text{Det}(S \parallel P) \mathbf{SC}^n P$.

Therefore, we consider each of the notions $\mathbf{FM}$, $\mathbf{KT}$ and $\mathbf{SC}^n$ as adequate when applied to a supervised plant. Note that these three are also known to be equivalent in the considered context.

7 Conclusion

In this paper, several notions of controllability of a supervisor w.r.t. a plant have been compared in different contexts and relations between these notions are mathematically characterized.

In the general context where supervisor and plant are nondeterministic, we gave an automata-theoretic characterization (**AC**) of controllability, and showed it is equivalent to language controllability. Of the existing controllability notions, in this context the equivalent notions of nondeterministic controllability by Flordal and Malik (**FM**-controllability) [10, 11] and Σ_u -admissibility by Kushi and Takai (**KT**-controllability) [16] are the only notions that imply language controllability.

From a practical perspective, it is more interesting to consider the controllability of the supervised plant w.r.t. the plant. We conclude that the notions of **FM**-controllability, Σ_u -admissibility, and (nondeterministic) state controllability [26] are equally appropriate for use when considering controllability of the supervised plant w.r.t. the plant.

So, when considering all contexts discussed in this paper, only **FM**-controllability and Σ_u -admissibility always imply language controllability.

The notion of observability is of similar importance in the field of supervisory control theory as controllability. Like controllability, it is used in different contexts, with different formalizations, and unexplored relationships between those. As future work, observability could be classified in a similar way.

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